Precipalm Prototype Documentation

Dynamic Stress-Based Nutrient Simulation Explanation

1. Overview of the Simulation

For this interactive prototype, the nutrient level values displayed upon map clicks are calculated using a **procedural simulation model** rather than a live machine learning inference server. This approach provides rapid, responsive visual feedback on the map and inside the PDF report without the overhead of heavy backend model computation for every click.

2. Mathematical Correlation Logic

The simulation is built on a direct mathematical correlation with the **NDVI (Normalized Difference Vegetation Index)** health proxy value generated dynamically by the Sentinel-2A simulation framework:

1. Stress Level Calculation:
The vegetative stress index is the inverse of the leaf greenness (NDVI):
Stress = Max(0.0, Min(1.0, 1.0 - NDVI))
• Healthy canopy (High NDVI ~ 0.9) produces Low Stress (~ 0.1).
• Stressed canopy (Low NDVI ~ 0.5) produces High Stress (~ 0.5).

2. Nutrient Value Calculations:
Leaf nutrient percentages (N, P, K, Mg) are calculated using linear regression mapping based on the stress index:
• **Nitrogen (N)**: *N = 2.65 - (Stress * 1.5)* (Target: 2.50%)
• **Phosphorus (P)**: *P = 0.165 - (Stress * 0.1)* (Target: 0.150%)
• **Potassium (K)**: *K = 0.98 - (Stress * 0.6)* (Target: 0.90%)
• **Magnesium (Mg)**: *Mg = 0.27 - (Stress * 0.16)* (Target: 0.250%)

3. Javascript Code Implementation

```
// Calculate local stress index from NDVI
const stress = Math.max(0.0, Math.min(1.0, 1.0 - ndvi));

// Predict foliar levels based on stress
const nVal = 2.65 - (stress * 1.5);
const pVal = 0.165 - (stress * 0.1);
const kVal = 0.98 - (stress * 0.6);
const mgVal = 0.27 - (stress * 0.16);
```

4. Production Integration Pathway

When moving this application to production, you can replace this procedural simulation block in the frontend with a live REST API call that feeds the 12-feature vector (consisting of raw Sentinel-2 bands and calculated vegetation indices) directly into your pre-trained Random Forest or Deep Learning inference models hosted on the server.